

PRACHI DUDHE

☎ (979) 721-2564 | ✉ pra24@tamu.edu | 🔗 LinkedIn | 📁 Portfolio | 📍 Open to Relocation

SUMMARY

MS in Data Science with 2+ years of experience building agentic AI systems, RAG pipelines, and evaluation frameworks. Hands-on with LLM orchestration, multi-mode routing, fine-tuning (LoRA/PEFT), and designing evals that measure real agent behavior in production. Known for diagnosing non-obvious agent failures and iterating quickly from real usage signals to measurable improvements

EDUCATION

Texas A&M University

Master of Science in Data Science (ECE Track)

Aug 2024 to May 2026

College Station, TX

Pune University

Bachelor of Engineering in Electronics and Telecommunication, Honors in Data Science

Aug 2018 to May 2022

India

PROFESSIONAL EXPERIENCE

Graduate Capstone – AgriChat | 🌐 Project

Texas A&M University - Corpus Christi

Aug 2025 to Dec 2025

Remote, Texas

- Built AgriChat end-to-end: designed a multi-mode agentic query router that switches between structured SQL execution, semantic RAG retrieval, and batch summarization based on inferred user intent – an early prototype of agents that learn what kind of answer a user actually needs
- Built offline evaluation harness measuring hallucination rate and answer groundedness against retrieved evidence; implemented structured output validation and rejection rules that logged malformed outputs rather than passing them silently downstream

Data Science Intern

Sulzer

Jun 2025 to Jul 2025

India

- Built a RAG document retrieval system (Python, embeddings, vector search) over 5,000+ technical maintenance records; raised retrieval accuracy by 31% and cut latency from 8.5s to 4.5s via semantic chunking, hybrid retrieval, and caching
- Designed modular, well-documented pipeline components with explicit optimization trade-offs to support team adoption and future iteration

Data Analyst

Tata Consultancy Services

Oct 2022 to Jul 2024

India

- Developed custom Infor 4GL extraction scripts within the ERP environment to process complex supply chain metrics; built an ETL pipeline (Oracle → SQL Server staging, nightly Agent jobs) and designed SQL views bridging the ERP's fragmented schema into Power BI dashboards tracking 10+ KPIs – cutting report preparation time by 25%
- Delivered a vendor spend dashboard against a hard deadline; built a normalisation mapping in Power Query to resolve duplicate records in the Business Partner master table, securing 95% user adoption

TECHNICAL PROJECTS

Road Lane Overlay: VLM-Guided Generative Restoration | Python, PyTorch, LoRA, Image Analysis, Computer Vision, Deep Learning

- Co-developed a restore-detect-repair pipeline recovering lane markings in degraded dashcam frames: dehazing (Dark Channel Prior), CLRRerNet lane detection over 62K CULane images, and guided inpainting confined to 6-12px lane-band gap masks
- Diagnosed why blind inpainting fails on lane geometry (LaMa fills gaps with asphalt texture rather than reconstructing thin high-contrast lines) and drove the switch to Qwen-Image-Edit fine-tuned with LoRA on 900 image pairs, improving FID by 12.3%, lane F1 by 7.1%, and recall by 10.3% over zero-shot

Telecom AI Issue Detection System | Python, scikit-learn, Random Forest, Time-Series, Anomaly Detection, Predictive Maintenance | 🐙 GitHub

- Designed and built a predictive maintenance and fault detection system across three modeling layers: KPI incident detection (Random Forest + SMOTE, 51% recall on 150K time-series rows), complaint signal detection (RoBERTa + zero-shot labeling, 77% recall on 50K records), and capacity utilization forecasting (Random Forest Regression, 14% MAE on 10M rows); deployed as a live Streamlit diagnostic dashboard for non-technical users
- Diagnosed four non-obvious agent and modeling failures: RCA agent reading historical averages instead of live KPIs (fixed by injecting live state and priority override logic), time-series data leakage (GroupShuffleSplit), majority-class trap causing 0% recall (class weighting), and 34% label noise from keyword heuristics (dual-signal transformer labeling)

CIFAR-10 (60K) Image Classification | Python, PyTorch, scikit-learn, Statistical Analysis

- Evaluated three architectures on CIFAR-10: baseline CNN (63%), hybrid ResNet-18 + SVM replacing the FC layer (86%), and ResNet-34 (95%)
- Applied saliency maps to investigate patterns underlying model predictions and validate that improvements reflected genuine signal rather than data artifacts

TECHNICAL SKILLS

- **Agentic AI & LLMs:** LangGraph, LlamaIndex, LangChain, RAG pipelines, agentic orchestration, prompt engineering, fine-tuning (LoRA/PEFT/QLoRA), LLM integration (Groq, Llama 3.3, Ollama), embedding-based search, structured output validation, prompt optimization
- **Evaluation & Research:** Eval harness design, offline evaluation, hallucination measurement, answer groundedness, production trace analysis, agent failure diagnosis, experiment design, statistical modeling, technical report writing
- **ML & Data Science:** Python (Pandas, NumPy, scikit-learn, PyTorch), SQL, predictive modeling, time-series analysis, anomaly detection, feature engineering, model evaluation, deep learning
- **Tools:** Git, Linux/Unix, Streamlit, Plotly, PowerBI

CERTIFICATIONS

- 5G Introductory Certification – Qualcomm Wireless Academy | 📜 Certificate